

Amazon

You are a Data Analyst on the Amazon Marketplace Analytics team focused on optimizing fee structures for third-party sellers....

Question 1: For each seller, please identify their top sale transaction in April 2024 based on sale amount. If there are multiple transactions with the same sale amount, select the one with the most recent sale_date.

```
with calculate AS(  
    SELECT *,  
        ROW_NUMBER() OVER(PARTITION BY seller_id ORDER BY sale_a  
mount DESC, sale_date DESC) AS row_num  
    FROM fct_seller_sales  
    WHERE sale_date >= '2024-04-01' and sale_date < '2024-05-01'  
    ORDER BY seller_id)  
  
SELECT c.seller_id, seller_name, sale_amount, fee_amount_perce  
ntage, sale_date  
FROM calculate AS c  
INNER JOIN dim_seller AS d  
ON c.seller_id = d.seller_id  
WHERE row_num = 1
```



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Question 2: Within May 2024, for each seller ID, please generate a weekly summary that reports the total number of sales transactions and shows the fee amount from the most recent sale in that week. This analysis will let us correlate fee changes with weekly seller performance trends.

```
WITH sales_may AS (  
    SELECT  
        s.seller_id,  
        d.seller_name,  
        TO_CHAR(sale_date, 'IYYY-IW') AS week_start,  
        s.sale_id,  
        s.sale_date,  
        s.sale_amount * s.fee_amount_percentage AS fee_amount  
    FROM fct_seller_sales s  
    JOIN dim_seller d ON s.seller_id = d.seller_id  
    WHERE s.sale_date >= DATE '2024-05-01' AND s.sale_date < DATE '2024-06-01'),  
  
    per_week AS(  
        SELECT seller_id,  
            seller_name,  
            week_start,  
            COUNT(sale_id) AS total_sale_transaction  
        FROM sales_may  
        GROUP BY seller_id, seller_name, week_start),  
  
    rank_sale AS(  
        SELECT *,  
            RANK() OVER (PARTITION BY seller_id, week_start ORDER  
                BY sale_date DESC, sale_id DESC) AS fee_rank  
        FROM sales_may)  
  
SELECT r.seller_id,  
    r.seller_name,  
    r.week_start,  
    total_sale_transaction,  
    fee_amount  
FROM rank_sale AS r  
INNER JOIN per_week AS p  
ON p.seller_id = r.seller_id  
    AND p.seller_name = r.seller_name  
    AND p.week_start = r.week_start  
WHERE fee_rank = 1
```



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Question 3: Using June 2024, for each seller, create a daily report that computes a cumulative count of transactions up to that day.

```
WITH sales_june AS (  
    SELECT  
        s.seller_id,  
        d.seller_name,  
        s.sale_date  
    FROM fct_seller_sales s  
    JOIN dim_seller d ON s.seller_id = d.seller_id  
    WHERE s.sale_date >= DATE '2024-06-01' AND s.sale_date < DATE '2024-07-01'  
    ORDER BY seller_id, seller_name, sale_date)  
  
SELECT  
    seller_id,  
    seller_name,  
    sale_date,  
    COUNT(*) OVER (  
        PARTITION BY seller_id  
        ORDER BY sale_date  
        ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW  
    ) AS cumulative_transaction_count  
FROM sales_june  
ORDER BY seller_id, sale_date;
```

